Statement of Purpose

I am currently pursuing a Bachelor of Technology in Computer Science and Engineering at the Indian Institute of Information Technology, Tiruchirappalli, with an expected graduation in 2027 and a CGPA of 7.7/10.0. My academic journey and hands-on technical experience have shaped a deep-seated interest in backend systems, scalable architectures, and high-performance software engineering. My fascination with computing began during my higher secondary education. However, it evolved into a professional pursuit during my undergraduate studies as I began to look past the syntax to understand the complex infrastructures powering modern digital applications. I have since built a rigorous foundation in Data Structures and Algorithms (DSA), Operating Systems, DBMS, Object-Oriented Programming, and System Design. Technical Excellence & Problem Solving I view competitive programming as the ultimate gym for analytical thinking. To date, I have solved over 300 DSA problems in C++, achieving a top 15% global ranking. This consistent practice has not only strengthened my problem-solving capabilities but has also taught me to write code where time and space complexity are treated as first-class constraints. Engineering Impact: CloudScaleX I believe that theory is best validated through engineering. I have developed multiple full-stack applications, integrating React frontends with Spring Boot microservices. In these projects, I implemented JWT-based stateless authentication and role-based access control to ensure secure, industry-standard API designs. My most significant architectural challenge was CloudScaleX, a microservices-based SaaS platform. I engineered this system using Spring Boot, PostgreSQL, Redis, and Docker. By identifying bottlenecks in data retrieval and implementing strategic Redis caching, I successfully reduced API latency by 30%. This project provided me with invaluable practical exposure to database optimization, caching strategies, and the importance of a clean, layered architecture in distributed systems. Leadership and Vision Beyond the IDE, I am committed to fostering a collaborative engineering culture. I have actively demonstrated leadership by mentoring my peers in C++ and DSA, helping to streamline our group's problem-solving efficiency. I believe that while individual code must be performant, it is strong communication and teamwork that allow an engineering organization to scale. My long-term goal is to contribute to the development of high-performance distributed systems that solve real-world problems at a global scale. I am a highly motivated and disciplined engineer, committed to the pursuit of technical depth and professional excellence.